RAG Learning

2.3 Sequences

OCR GCSE Mathematics (J560) — Paper 1, 2 & 3 — Foundation and Higher

2.3.1 Describing Sequences

A **sequence** is an ordered list of numbers (or objects) following a rule. Each number is a **term**, labelled $T_1, T_2, T_3, \ldots$ or $u_1, u_2, u_3, \ldots$

A **term-to-term rule** describes how to get from one term to the next.
A **position-to-term rule** (the nth term formula) gives any term directly from its position n.

Arithmetic (Linear) Sequences

Each term differs from the previous by a fixed amount called the **common difference** d.

- If $d > 0$: the sequence is increasing.
- If $d < 0$: the sequence is decreasing.

Sequence	First differences	Common difference d	Type
$3, 7, 11, 15, 19, \ldots$	$+4, +4, +4, +4$	4	Arithmetic
$20, 17, 14, 11, \ldots$	$-3, -3, -3$	-3	Arithmetic
$1, 4, 9, 16, 25, \ldots$	$+3, +5, +7, +9$	Not constant	Not arithmetic

Geometric Sequences

Each term is obtained by **multiplying** the previous term by a fixed value called the **common ratio** r.

Sequence	Ratios	Common ratio r
$2, 6, 18, 54, 162, \ldots$	$\times 3, \times 3, \times 3, \times 3$	3
$80, 40, 20, 10, 5, \ldots$	$\times 0.5, \times 0.5, \ldots$	0.5
$1, -2, 4, -8, 16, \ldots$	$\times(-2), \times(-2), \ldots$	-2

Exam Tip: To identify the type of sequence: check first differences. If they are constant, the sequence is arithmetic. If not constant, check second differences. If *those* are constant, it is quadratic. If terms have a constant ratio, it is geometric.

2.3.2 The nth Term of Arithmetic Sequences

For an arithmetic sequence with first term a and common difference d:

$$T_n = a + (n-1)d$$

This can always be written in the form $T_n = dn + c$, where d is the common difference and $c = a - d$.

Finding the nth Term Formula

Method:

1. Find the common difference d — this is the coefficient of n.
2. Work out what $d \times 1$ gives and adjust to match the first term. The adjustment is $c = T_1 - d$.
3. Write the formula: $T_n = dn + c$.

Worked Example: Find the nth term of $5, 8, 11, 14, \ldots$

> Common difference: $d = 3$
>
> Using $n = 1$: $3(1) + c = 5 \Rightarrow c = 2$

> nth term: $T_n = 3n + 2$
>
> Check: $T_3 = 3(3) + 2 = 11$ ✓

Worked Example: Find the nth term of $17, 13, 9, 5, \ldots$

> $d = -4$
>
> $-4(1) + c = 17 \Rightarrow c = 21$
>
> $T_n = -4n + 21$
>
> Check: $T_4 = -16 + 21 = 5$ ✓

Using the Formula

Worked Example: Is 100 a term in the sequence $7, 11, 15, 19, \ldots$?

> nth term: $T_n = 4n + 3$
>
> Set equal to 100: $4n + 3 = 100 \Rightarrow 4n = 97 \Rightarrow n = 24.25$
>
> Not a whole number, so 100 is **not** a term in this sequence.

Worked Example: The nth term of a sequence is $5n - 3$. Which term equals 72?

> $5n - 3 = 72 \Rightarrow 5n = 75 \Rightarrow n = 15$
>
> 72 is the **15th term**.

Exam Tip: To check whether a value is in a sequence, set T_n equal to that value and solve for n. If n is a positive integer, the value is in the sequence.

2.3.3 Quadratic Sequences [Higher]

Higher only

A **quadratic sequence** has a constant **second difference**. The nth term formula contains n^2 as the highest power, i.e. $T_n = an^2 + bn + c$.

Identifying Quadratic Sequences Using Differences

Position n	1	2	3	4	5
Term T_n	3	8	15	24	35
1st differences		5	7	9	11
2nd differences			2	2	2

Second differences are constant at 2, confirming this is quadratic.

Finding the nth Term of a Quadratic Sequence

Method:

1. Find the second difference. The coefficient of n^2 is $a = \dfrac{\text{second difference}}{2}$.
2. Subtract the an^2 sequence from the original to get a *linear* remainder sequence.
3. Find the nth term of the remainder (it will be a linear expression $bn + c$).
4. Write the complete formula: $T_n = an^2 + bn + c$.

Worked Example: Find the nth term of $3, 8, 15, 24, 35, \ldots$

> Second difference = 2, so $a = 1$. The n^2 sequence is $1, 4, 9, 16, 25, \ldots$
>
> Subtract: $3 - 1 = 2,\ 8 - 4 = 4,\ 15 - 9 = 6,\ 24 - 16 = 8,\ 35 - 25 = 10$
>
> Remainder sequence: $2, 4, 6, 8, 10, \ldots$ This is linear with nth term $2n$.

Complete formula: $T_n = n^2 + 2n$

Check: $T_3 = 9 + 6 = 15$ ✓, $T_5 = 25 + 10 = 35$ ✓

Worked Example: Find the nth term of $5, 14, 27, 44, 65, \ldots$

1st differences: $9, 13, 17, 21$ — 2nd differences: $4, 4, 4$ — constant, so quadratic.

$a = 4/2 = 2$. The $2n^2$ sequence: $2, 8, 18, 32, 50$

Subtract: $5 - 2 = 3,\ 14 - 8 = 6,\ 27 - 18 = 9,\ 44 - 32 = 12,\ 65 - 50 = 15$

Remainder: $3, 6, 9, 12, 15, \ldots$ → nth term = $3n$

Complete: $T_n = 2n^2 + 3n$

Check: $T_2 = 8 + 6 = 14$ ✓

Exam Tip: For quadratic sequences: always calculate BOTH the first AND second differences in your working. Examiners award marks for this even if you make an arithmetic error later.

Summary Comparison: Arithmetic vs Quadratic Sequences

Feature	Arithmetic (linear)	Quadratic
1st differences	Constant	Not constant (but increasing by fixed amount)
2nd differences	Zero	Constant (non-zero)
nth term form	$dn + c$	$an^2 + bn + c$
Coefficient of n^2	0	$a = \frac{\text{2nd difference}}{2}$
Graph of T_n vs n	Straight line	Parabola (curve)

2.3.4 Special Sequences

Square Numbers

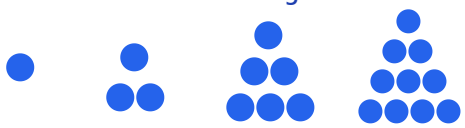
$1, 4, 9, 16, 25, 36, 49, 64, 81, 100, \ldots \quad T_n = n^2$

First differences: $3, 5, 7, 9, 11, \ldots$ (odd numbers); Second differences: all 2.

Triangular Numbers

$1, 3, 6, 10, 15, 21, 28, 36, \ldots \quad T_n = \dfrac{n(n+1)}{2}$

Triangular numbers as dot arrays



Cube Numbers

$1, 8, 27, 64, 125, 216, \ldots \quad T_n = n^3$

Powers of 2 (Geometric)

$1, 2, 4, 8, 16, 32, 64, \ldots \quad T_n = 2^{n-1}$, common ratio $r = 2$.

Fibonacci Sequence

$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$

Term-to-term rule: $T_{n+1} = T_n + T_{n-1}$ (each term is the sum of the two preceding terms). There is no simple closed-form nth term formula at GCSE level.

- Fibonacci numbers appear in nature — spiral patterns on sunflowers, pinecones, and shells.
- The ratio of consecutive Fibonacci terms approaches the golden ratio $\phi \approx 1.618$.

Other Common Special Sequences

Name	Sequence	nth term
Natural numbers	$1, 2, 3, 4, 5, \ldots$	n
Even numbers	$2, 4, 6, 8, 10, \ldots$	$2n$
Odd numbers	$1, 3, 5, 7, 9, \ldots$	$2n - 1$
Square numbers	$1, 4, 9, 16, 25, \ldots$	n^2
Triangular numbers	$1, 3, 6, 10, 15, \ldots$	$\frac{n(n+1)}{2}$
Cube numbers	$1, 8, 27, 64, 125, \ldots$	n^3
Powers of 2	$2, 4, 8, 16, 32, \ldots$	2^n

Worked Example: Show that 210 is a triangular number and find its position.

> Set $\frac{n(n+1)}{2} = 210$
>
> $n(n+1) = 420$
>
> Test: $n = 20$: $20 \times 21 = 420$ ✓
>
> So 210 is the **20th** triangular number.

Worked Example: A geometric sequence has first term 5 and common ratio 3. Write the first five terms and find the nth term.

> Terms: $5, 15, 45, 135, 405$
>
> nth term: $T_n = 5 \times 3^{n-1}$
>
> Check: $T_3 = 5 \times 3^2 = 5 \times 9 = 45$ ✓

Exam Tip: You may be asked to recognise a sequence as triangular, square, or Fibonacci — memorise these! Also be able to state the nth term formula for each.

2.3.5 Recurrence Relations [A-Level]

A **recurrence relation** defines each term using one or more previous terms. The general form is:

$$u_{n+1} = f(u_n)$$

together with an **initial condition** (the first term u_1, or sometimes u_0).

For example, $u_{n+1} = 2u_n + 3$ with $u_1 = 1$ generates the sequence $1, 5, 13, 29, \ldots$ — each term is found by doubling the previous term and adding 3.

Worked Example: A sequence is defined by $u_{n+1} = \dfrac{u_n + 6}{u_n + 1}$, with $u_1 = 2$.

Find u_2, u_3, and u_4, giving values to 3 decimal places where necessary.

$$u_2 = \frac{2+6}{2+1} = \frac{8}{3} \approx 2.667$$

$$u_3 = \frac{8/3+6}{8/3+1} = \frac{26/3}{11/3} = \frac{26}{11} \approx 2.364$$

$$u_4 = \frac{26/11+6}{26/11+1} = \frac{92/11}{37/11} = \frac{92}{37} \approx 2.486$$

The sequence appears to be converging towards a limit.

Finding a Fixed Point (Limit of a Convergent Recurrence)

If the sequence converges to a limit L, then as $n \to \infty$, both u_{n+1} and u_n approach L. Substitute $u_{n+1} = u_n = L$ into the recurrence relation and solve.

Worked Example: Find the limit of the sequence defined by $u_{n+1} = \dfrac{u_n + 6}{u_n + 1}$.

Set $u_{n+1} = u_n = L$: $L = \dfrac{L+6}{L+1}$

$L(L+1) = L + 6 \Rightarrow L^2 + L = L + 6 \Rightarrow L^2 = 6 \Rightarrow L = \sqrt{6} \approx 2.449$

(Taking the positive root since $u_1 = 2 > 0$.)

Exam Tip: When finding the limit of a recurrence relation, replace *both* u_{n+1} and u_n with L, then solve the resulting equation. Check the sign or value is consistent with the initial terms. A cobweb or staircase diagram (not required but helpful) shows graphically whether the sequence converges or diverges.

2.3.6 Behaviour of Sequences [A-Level]

Sequences can be classified by their long-term behaviour:

- **Increasing**: $u_{n+1} > u_n$ for all n
- **Decreasing**: $u_{n+1} < u_n$ for all n
- **Convergent**: terms approach a fixed limit L as $n \to \infty$
- **Divergent**: terms grow without bound (e.g. geometric with $|r| > 1$)
- **Periodic**: the sequence repeats after a fixed number of terms (the **period**)
- **Oscillating**: terms alternate between two or more values, often converging to a limit (e.g. geometric with $-1 < r < 0$) or diverging (e.g. $r < -1$)

Geometric ratio r	Behaviour	Example ($a = 2$)
$r > 1$	Diverges (increases without bound)	$2, 4, 8, 16, \ldots$ ($r = 2$)
$0 < r < 1$	Converges to 0	$2, 1, 0.5, 0.25, \ldots$ ($r = 0.5$)

$r = 1$	Constant (periodic with period 1)	$2, 2, 2, 2, \ldots$
$-1 < r < 0$	Oscillates and converges to 0	$2, -1, 0.5, -0.25, \ldots$ ($r = -0.5$)
$r = -1$	Periodic with period 2	$2, -2, 2, -2, \ldots$
$r < -1$	Oscillates and diverges	$2, -4, 8, -16, \ldots$ ($r = -2$)

Worked Example: The sequence defined by $u_{n+1} = -u_n + 3$, $u_1 = 1$.

> $u_1 = 1,\ u_2 = 2,\ u_3 = 1,\ u_4 = 2,\ u_5 = 1, \ldots$
>
> This is a **periodic sequence** with period 2. It oscillates between 1 and 2 indefinitely.
>
> Fixed point: $L = -L + 3 \Rightarrow 2L = 3 \Rightarrow L = 1.5$ — the sequence oscillates around 1.5 but never reaches it.

2.3.7 Sigma Notation [A-Level]

The Greek letter Σ (sigma) is used to write a sum compactly.

$$\sum_{r=1}^{n} u_r = u_1 + u_2 + u_3 + \cdots + u_n$$

The variable r is the index of summation, running from the lower limit to the upper limit.

Sigma expression	Expanded form	Sum
$\sum_{r=1}^{4} r$	$1 + 2 + 3 + 4$	10
$\sum_{r=1}^{5} 2r$	$2 + 4 + 6 + 8 + 10$	30

$\sum_{r=1}^{n} r$	$1+2+\cdots+n$	$\frac{n(n+1)}{2}$
$\sum_{r=1}^{n} c$	$c+c+\cdots+c$ (n times)	cn

Worked Example: Evaluate $\sum_{r=3}^{7}(2r-1)$.

Write out the terms: $r=3:5;\ r=4:7;\ r=5:9;\ r=6:11;\ r=7:13$

Sum $=5+7+9+11+13=45$

Alternatively: $\sum_{r=3}^{7}(2r-1)=\sum_{r=1}^{7}(2r-1)-\sum_{r=1}^{2}(2r-1)=49-4=45$

Exam Tip: To evaluate $\sum_{r=k}^{n}$, split it as $\sum_{r=1}^{n}-\sum_{r=1}^{k-1}$. This uses the standard results (see below) for sums starting at $r=1$.

2.3.8 Sum of an Arithmetic Series [A-Level]

The sum of the first n terms of an arithmetic series with first term a and common difference d:

$$S_n=\frac{n}{2}(2a+(n-1)d)$$

Equivalently, if the last term is $l=a+(n-1)d$:

$$S_n=\frac{n}{2}(a+l)$$

Derivation

Write the sum forwards and backwards, then add:

$$S_n = a + (a+d) + (a+2d) + \cdots + l$$

$$S_n = l + (l-d) + (l-2d) + \cdots + a$$

$$2S_n = n(a+l) \implies S_n = \frac{n}{2}(a+l)$$

Worked Example: Find the sum of the first 20 terms of the arithmetic series $3 + 7 + 11 + 15 + \ldots$

> $a = 3$, $d = 4$, $n = 20$
>
> $S_{20} = \frac{20}{2}(2 \times 3 + 19 \times 4) = 10(6 + 76) = 10 \times 82 = 820$

Worked Example: The sum of the first n terms of an arithmetic series is $S_n = 3n^2 + 2n$. Find the first term and common difference.

> $T_1 = S_1 = 3 + 2 = 5$, so $a = 5$
>
> $T_2 = S_2 - S_1 = (12 + 4) - 5 = 11$
>
> $d = T_2 - T_1 = 11 - 5 = 6$
>
> Check: $T_n = S_n - S_{n-1} = (3n^2 + 2n) - (3(n-1)^2 + 2(n-1)) = 6n - 1$ ✓ (gives $T_1 = 5, T_2 = 11$)

2.3.9 Sum of a Geometric Series [A-Level]

The sum of the first n terms of a geometric series with first term a and common ratio r ($r \neq 1$):

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n - 1)}{r - 1}$$

(Use whichever form avoids a negative denominator in context.)

Derivation

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1}$$

$$rS_n = ar + ar^2 + \cdots + ar^{n-1} + ar^n$$

$$S_n - rS_n = a - ar^n \implies S_n(1-r) = a(1-r^n) \implies S_n = \frac{a(1-r^n)}{1-r}$$

Worked Example: Find the sum of the first 8 terms of $2 + 6 + 18 + 54 + \ldots$

> $a = 2$, $r = 3$, $n = 8$
>
> $S_8 = \dfrac{2(3^8 - 1)}{3 - 1} = \dfrac{2(6561 - 1)}{2} = 6560$

Worked Example: A geometric series has first term 5 and sum to 6 terms equal to 315. Find the common ratio.

> $\dfrac{5(r^6 - 1)}{r - 1} = 315 \Rightarrow \dfrac{r^6 - 1}{r - 1} = 63$
>
> Try $r = 2$: $\dfrac{64 - 1}{1} = 63$ ✓ So $r = 2$.

2.3.10 Sum to Infinity of a Geometric Series [A-Level]

When $|r| < 1$, the terms of a geometric series get smaller and the sum approaches a finite limit as $n \to \infty$:

$$S_\infty = \frac{a}{1-r} \quad \text{provided } |r| < 1$$

If $|r| \geq 1$, the series diverges and no sum to infinity exists.

Why the Formula Works

From $S_n = \dfrac{a(1-r^n)}{1-r}$: when $|r| < 1$, $r^n \to 0$ as $n \to \infty$, so

$$S_\infty = \lim_{n\to\infty} S_n = \frac{a(1-0)}{1-r} = \frac{a}{1-r}$$

Worked Example: Find the sum to infinity of $4 + 2 + 1 + 0.5 + \ldots$

> $a = 4$, $r = 0.5$ (since $|r| < 1$, the sum to infinity exists)
>
> $S_\infty = \dfrac{4}{1-0.5} = \dfrac{4}{0.5} = 8$

Worked Example: A geometric series has first term a and common ratio r. The second term is 4 and the sum to infinity is 25. Find a and r.

> Second term: $ar = 4$, so $a = \dfrac{4}{r}$
>
> Sum to infinity: $\dfrac{a}{1-r} = 25$
>
> Substitute: $\dfrac{4/r}{1-r} = 25 \Rightarrow 4 = 25r(1-r) = 25r - 25r^2$
>
> $25r^2 - 25r + 4 = 0 \Rightarrow (5r-1)(5r-4) = 0 \Rightarrow r = \dfrac{1}{5}$ or $r = \dfrac{4}{5}$
>
> Both satisfy $|r| < 1$. If $r = \frac{1}{5}$: $a = 20$. If $r = \frac{4}{5}$: $a = 5$.

Exam Tip: Always check $|r| < 1$ before claiming a sum to infinity exists. If the question doesn't tell you the series converges, check the ratio first. For the sum to n terms, use $S_n = \dfrac{a(1-r^n)}{1-r}$; for the sum to infinity use $S_\infty = \dfrac{a}{1-r}$.

Quick Reference Summary

Concept	Key fact / formula

Arithmetic sequence	Constant 1st differences; $T_n = dn + c$
Geometric sequence	Constant ratio; $T_n = a \cdot r^{n-1}$
Finding nth term (linear)	d = coefficient of n; $c = T_1 - d$
Is k in sequence?	Solve $T_n = k$; n must be a positive integer
Quadratic sequence [H]	Constant 2nd differences; $a = \frac{\text{2nd diff}}{2}$; subtract an^2 and solve remainder
Triangular numbers	$T_n = \frac{n(n+1)}{2}$: 1, 3, 6, 10, 15 ...
Square numbers	$T_n = n^2$: 1, 4, 9, 16, 25 ...
Fibonacci rule	Each term = sum of the two before it
Recurrence relation [AL]	$u_{n+1} = f(u_n)$; find limit L by setting $u_{n+1} = u_n = L$
Sum of arithmetic series [AL]	$S_n = \frac{n}{2}(2a + (n-1)d)$
Sum of geometric series [AL]	$S_n = \frac{a(1-r^n)}{1-r} \quad (r \neq 1)$
Sum to infinity [AL]	$S_\infty = \frac{a}{1-r}$, only if $\lvert r \rvert < 1$
Periodic/oscillating [AL]	Geometric: periodic if $r = -1$; oscillating if $r < 0$